

HW#7; Problems Statement:

10-4 **D** For the circuit of Figure P10-4:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- Select values of R and C to locate a zero at -33 krad/s.

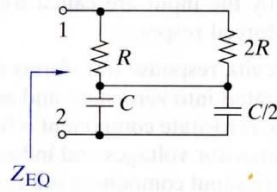


FIGURE P10-4

10-6 **D** For the circuit of Figure P10-6:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- Select values of R and L to locate a pole at -150 rad/s. Where is the resulting zero?

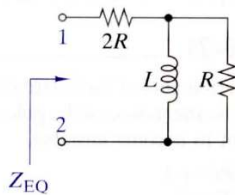


FIGURE P10-6

10-8 **D** For the circuit of Figure P10-8:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- If $R = 10$ k Ω , select values of L and C to locate poles at $\pm j200$ krad/s. Where are the resulting zeros?

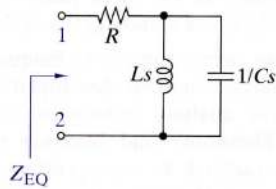


FIGURE P10-8

10-17 **D** Find the Thévenin equivalent for the circuit in Figure P10-17. Then select values for R and L so that the Thévenin voltage has a pole at -12 krad/s.

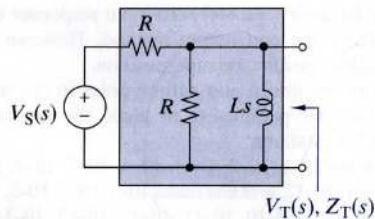


FIGURE P10-17

- 10-21 The switch in Figure P10-21 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$, $i_C(t)$, $V_O(s)$, and $v_O(t)$ in symbolic form.

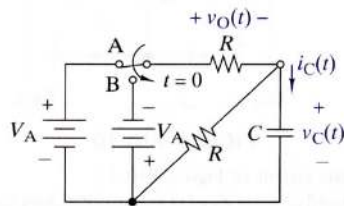


FIGURE P10-21

- 10-24 Transform the circuit in Figure P10-23 into the s domain and find $I_L(s)$ and $i_L(t)$ when $v_1(t) = V_A e^{-1000t} u(t)$, $R = 200 \Omega$, $L = 100 \text{ mH}$, and $i_L(0) = 0 \text{ A}$.

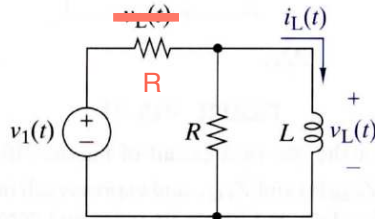


FIGURE P10-23

- 10-28 The circuit in Figure P10-28 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $V_O(s)$. Identify the poles and the zeros.

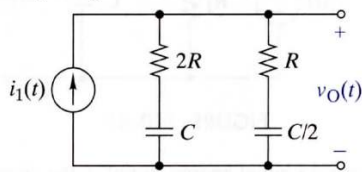


FIGURE P10-28

- 10-33 Repeat Problem 10-32 when $i_S(t) = 2.5 \cos 2000t u(t) \text{ mA}$.

- 10-32 There is no energy stored in the capacitor in Figure P10-32 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 5 e^{-500t} u(t) \text{ mA}$. Identify the forced and natural poles in $V_O(s)$.

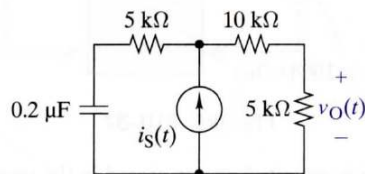


FIGURE P10-32

- 10-53 **D** The OP AMP circuit in Figure P10-53 is in the zero state. Use node-voltage equations to find the circuit determinant. Select values of R , C_1 , and C_2 so that the circuit has $\omega_0 = 10 \text{ krad/s}$ and $\zeta = 1$.

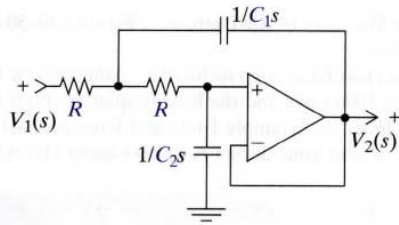


FIGURE P10-53

- 10-68 **s-domain OP AMP Circuit Analysis** **A**

The OP AMP circuit in Figure P10-68 is in the zero-state. Transform the circuit into the s -domain and use the OP AMP circuit analysis techniques developed in Section 4-4 to find the relationship between the input $V_1(s)$ and the output $V_2(s)$.

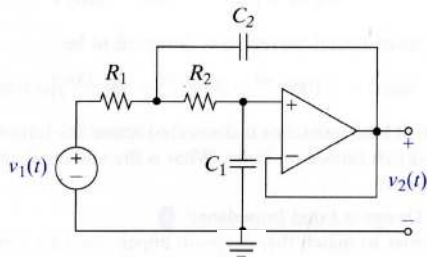


FIGURE P10-68